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Paleomagnetism and potassium-argon age of the Messejana Dike (Portugal and Spain): angular limitation to the rotation of the Iberian Peninsula since the Middle Jurassic

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A paleomagnetic and potassium-argon dating investigation has been carried out on a 530-km-long dike system which transects the western Iberian Peninsula in a northeasterly direction. The K-Ar age determinations were made on mineral separates exclusively. They range between 160 and 200 Ma and the authors suppose that this reflects the actual time interval of the intrusion, in accord with previous results. The paleomagnetic pole derived from 12 sites regularly distributed along the dike (71°N, 236°E) coincides well with other Mesozoic paleomagnetic poles from the western Africa. A contemporaneous pole from stable Europe is tentatively deduced from African and North American Late Triassic/Early Jurassic poles using different reconstruction models around the North Atlantic Ocean. The divergence between this pole and the Iberian pole corresponds to the result obtained for Permian poles.

1. Introduction

The Messejana dike of tholeiitic diabase stretches northeasterly over a distance of 530 km, from Odemira on the Atlantic coast in Portugal to Avila in Spain. It is related to the basaltic igneous activity known in the continental zones bordering the central Atlantic Ocean of Late Triassic/Early Jurassic age. The lava flows and intrusive rocks representing the magmatic activity were the subject of many paleomagnetic and geochronologic studies, in western Africa [1,2], Morocco [3,4] and mainly in the eastern Appalachians [5–7]. The authors associate this magmatic activity with the separation of Africa from North America. It marks the initiation of rifting or predates it by several tens of millions of years. Occurrences of such Mesozoic diabases are scarce in Europe with the exception of the Iberian Peninsula where the Messejana dike fault system represents a conspicuous lineament. No other dolerite dike in the zones around the North Atlantic approaches the dimension of the Messejana dike except for the Foum Zguid dike in South Morocco.

The objectives of the investigation were to provide geochronologic and paleomagnetic information for a period where the data are rather scarce concerning the Iberian Peninsula as well as Europe. They are useful to elucidate the chronology of the rotation of Iberia relative to Europe and to assess its magnitude.

2. Geological setting

The previous geological and geochronological investigations carried out on the Messejana fault-dike system have recently been reviewed [8]. The wrench fault affects Paleozoic or even older formations of the Iberian Meseta. Unmetamorphosed Tertiary sedimentary rocks overlie the extremities and also the Messejana Aljustrel part of the fault dike system. This system is the most important tectonic feature among those striking NW-SE or SW-NE [9], associated with minor faults striking NNE-SSW. The earliest observable offsets date

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from the Late Hercynian period. Later this set of fractures was remobilized, giving rise to magmatic intrusions and vertical motions. These tectonic events are either Mesozoic or Cenozoic.

Field geologists [8] estimate the left-lateral strike-slip of the Messejana fault to date from between Carboniferous and Triassic times. The offset amounts to 30 km [9], but other field observations [8,10,11] ascribe an insignificant dimension of 3 km to the displacement.

The fault has undergone several stages of movement. Three episodes of tectonic activity have been distinguished [12] in the Aljustrel-Messejana part of the system: (1) wrench faulting initiating the Messejana fracture; (2) tension fissuring allowing the ascent of the tholeiitic diabase; and (3) localized normal faulting in Tertiary times.

The fracture has been studied in detail [10] over 90 km between Albuquerque and Canaveral. The main observable displacement along the fault is of strike-slip nature with an offset comprised of between 2.5 and 3 km. Furthermore, a localized fracturation occurred subsequently to the emplacement of the tholeiitic diabase.

A thorough tectonic investigation carried out [13] on the Spanish part of the dike fault system underscores the importance of the Tertiary normal-faulting event. These movements have given rise to the central Spanish Mountains (Sierra de Gredos, Sierra de Guadarrama) and also to small sedimentary basins filled with Tertiary detrital material. The Spanish portion of the dike has been extensively studied, mainly from a petrological point of view, by Garcia de Figuerola [14,15], Garcia de Figuerola and Carnicero [16], and Garcia de Figuerola et al. [17]. Its Portuguese part was examined by Torre de Assunçao [18–21].

The intrusion of the diabase dike has accompanied the tension-fissuring movement. The duration of such a tensional phase can be assessed by use of one of the magmatic events indicated by the K-Ar determination performed on samples taken along the fault. Reliable data [8,22] give a range of 156–220 Ma. The large spread of the analytical data is regarded by Schermerhorn et al. [8] as reflecting the actual duration of the magmatic event. In contrast, the value of 220 Ma measured on plagioclase is considered by Teixeira and Torquato [22] as the real age of the dike emplacement; all other K-Ar dates could be explained by Ar diffusion loss.

3. Sampling and analytical techniques

Samples were collected in June 1975 for paleomagnetic study and K-Ar dating. The twelve investigated sites were regularly distributed over the dike fault system (Fig. 1). In so far as possible, sampling was carried out in fresh outcrops, mainly road trenches and quarries. From four to eleven hand samples were collected at each site. Orientation was performed with magnetic and sun compasses. Four to five specimens, either as 43-mm cubes, or as cores 25 mm in diameter and 25 mm long, were obtained from one sample. From all sites but one, a 10-kg piece of rock was taken for K-Ar dating.

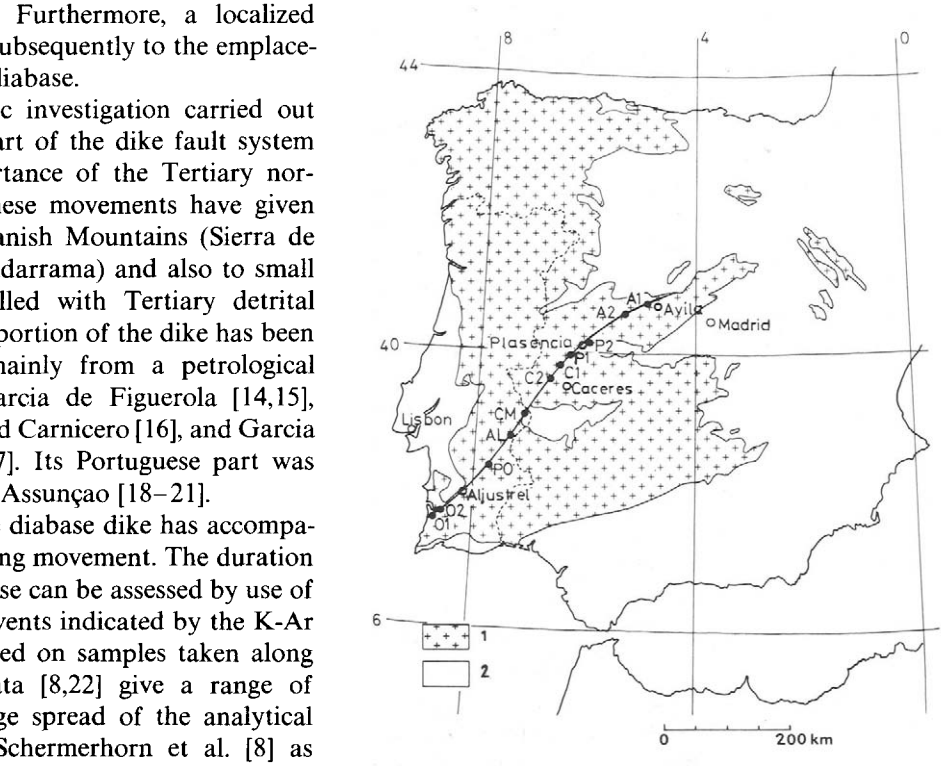


Fig. 1. Map of the Iberian Peninsula and location of sampling sites. *1* = Paleozoic of the Iberian Meseta; *2* = Mesozoic and Recent cover.

3.1. Geochronology

Plagioclases and in a few cases pyroxenes were analysed in this study. Whole-rock analyses were avoided for the following reasons:

— The samples often display incipient development of alteration products such as sericite in part of the plagioclase.

— The previous attempts to date the dike [8, 22–24], with the exception of one, were all performed on whole rock and displayed highly scattered absolute ages ranging from 156 to 275 Ma.

— Dalrymple and Lanphere [25], and Dalrymple et al. [2] showed that fresh plagioclases from volcanic rocks may yield consistent K-Ar dates. Westphal et al. [4] obtained K-Ar results more compatible with the geological record than those performed on whole rocks [26] when using plagioclase separates from Moroccan tholeiitic diabases.

Plagioclases and pyroxenes were obtained from the 100- to 160-μm sieve fraction by use of heavy liquids and magnetic separation. The analytical procedure, more extensively described elsewhere [4], was the following: potassium was measured by flame photometry with lithium internal standard [27,28]. Argon was extracted in a bakeable glass vacuum apparatus and determined by isotope dilution techniques (using ^{38}Ar as a tracer) in a MS 10S mass spectrometer. All samples were treated by the static method.

Two sets of constants were used to calculate ages (Table 1). The first one [25] was kept for the sake of convenience in order to compare directly our results with the already published K-Ar age determinations performed on the same dike. Plus-minus figures were estimates of analytical precision at one standard deviation. They were calculated following the procedure given by Cox and Dalrymple [29].

3.2. Paleomagnetism

The natural remanent magnetization (NRM) of the specimens was measured either with a spinner magnetometer (PAR) or a "Digico" fluxgate magnetometer. The following experiments were carried out on two or three specimens for each sample: (1) viscous magnetization evaluation by using the Thellier and Thellier [30] reversal method in the laboratory field, (2) chemical cleaning for about ten days with 6*N* hydrochloric acid, and (3) stepwise demagnetization comprising about ten steps in an alternating field of up to 38 or 56 mT.

Furthermore, the low field susceptibility was measured as a function of the temperature for seven sites uniformly distributed along the dike. The curves were obtained by means of a commercial induction bridge (Highmoor) that yields the susceptibility in an alternating field of 1.5 kHz and 0.25 mT. The most magnetic fraction of the rocks separated with an isodynamic Frantz device was used for this measurement. Finally, we determined the anisotropy of susceptibility on two to eight specimens for each site using cores 25 mm in diameter and 22.5 mm long, which had not been demagnetized, but only cleaned with 6*N* HCl. The susceptibility and anisotropy of susceptibility measurements were performed with a commercial a.c. bridge (Digico).

4. Age of the Messejana dike

The results of K-Ar age analyses on eleven plagioclases and three pyroxenes are given in Table 1.

The apparent age yielded by the plagioclase samples ranges from 130 to 206 Ma. It is not likely that the former date represents the age of crystallization since the corresponding pyroxene displays a value of 157 Ma. Freire d'Andrade and Schermerhorn [31], and Schermerhorn et al. [8] mention important Neogene faulting in the southwestern part of the dike. This tectonic event might have slightly disturbed the K-Ar system of plagioclase O1. Pyroxene and plagioclase of C2 and CM samples give within error margins similar apparent K-Ar ages. This convergence strongly suggests that the 170-Ma value attests to an intrusion date. To assess the meaning of the other plagioclase data we must call upon field arguments and previous K-Ar determinations both on the Messejana dike and on other Mesozoic tholeiitic diabases in Africa. Field observations in the Gavio region give evi-

TABLE 1

Analytical data and potassium-argon ages. The ± figures assigned to each age are estimates of analytical precision at one standard deviation

Sample No.	Mineral	K_2O (wt.%)	$^{40}Ar_{rad.}$ (10^{-11} mol/g)	100 $^{40}Ar_{rad.}/^{40}Ar_{total}$	Calculated age ± 1 s.d. (Ma) 1 *	2 **
A1	plagioclase	0.680	21.72	80	205±6	209±6
A2	plagioclase	3.790	105.3	95	179±5	183±5
P3	plagioclase	0.198	5.63	41	183±7	187±7
P2	plagioclase	0.405	10.24	77	164±5	168±5
P1	plagioclase	1.400	33.89	94	157±4	161±4
C2	plagioclase	1.070	28.24	72	171±5	175±5
C2	plagioclase $d>2.71$	0.910	23.74	95	169±5	173±5
C2	pyroxene	0.083	2.090	29	163±9	167±9
C1	plagioclase	0.740 0.750	22.62	83	195±5	199±5
CM	plagioclase	1.010	26.64	84	170±5	175±5
CM	pyroxene	0.087	2.144	49	160±9	164±9
AL	plagioclase	1.220	35.56	91	188±5	192±5
PO	plagioclase	1.730 1.750	52.25	96	193±5	197±5
OI	plagioclase	0.860 0.870	17.76	82	134±4	137±4
OI	pyroxene	0.079	1.914	57	157±9	161±9

* $\lambda_\epsilon = 0.585\times10^{-10}$ a^{-1}; $\lambda_\beta = 4.72\times10^{-10}$ a^{-1}; $^{40}K/K_{total} = 1.19\times10^{-4}$ mol/mol.
** $\lambda_\epsilon = 0.581\times10^{-10}$ a^{-1}; $\lambda_\beta = 4.962\times10^{-10}$ a^{-1}; $^{40}K/K_{total} = 1.167\times10^{-4}$ mol/mol.
d = density.

dence of multiple intrusions such as internal chilled contacts [8]. The corresponding K-Ar ages in the Aljustrel-Messejana part of the dike fall in the range of 160–190 Ma. Furthermore, a review of the earlier geochronological investigations have led Schermerhorn et al. [8] to assign an age of intrusion between 160 and 200 Ma. This time span coincides with our results. If one looks at K-Ar data on tholeiitic diabase sills and flows from Morocco [4] once again the same time span of 40 Ma is found. Other examples include the dike swarms parallel to the Atlantic coast in Liberia that display an age range of 173–192 Ma, likewise interpreted as a multiple intrusion system [2], and the basalt sills of Namibia [32] with a time span of eruption of between 161 and 199 Ma.

In summary, we conclude that the K-Ar dates of the Messejana dike fall within the range of 160–200 Ma, dating the intrusion from the Late Triassic to the Middle Jurassic [33]. This range most probably points to a multiple intrusion system.

5. Paleomagnetism of the dike

5.1. *Viscous magnetization*

The Thellier and Thellier reversal test showed that the proportion of VRM acquired within a time span from 2 to 3 weeks was insignificant for most of the specimens (Table 2).

5.2. *Drilling and sawing magnetization*

Secondary magnetization was acquired upon core drilling and cube sawing. Our observations were similar to those already published [34,35]. Three features can be pointed out.

— In cores the magnetization was almost parallel to the axis. In cubes it displayed more scattering due to the effects produced on six sides.

— The intensity of the parasitic magnetization was high, reaching from 2 to 5 times the NRM intensity in a few cores.

—The influence of an alternating field on the parasitic magnetization and on the NRM was similar. Therefore, the occurrence of such a parasitic magnetization might have hampered the determination of the characteristic one. The parasitic magnetization was superficial and could be easily removed by concentrated HCl cleaning as suggested by Burmester [34]. It was necessary for the HCl to penetrate 1 or 1.5 mm in order to ensure a complete removal of sawing and coring magnetization. Thus the procedure had to last ten days. The magnetization directions (Figs. 2 and 3) after HCl treatment clustered nicely. However, this treatment caused a decrease in NRM and in the susceptibility of the specimens. The susceptibility decreased by 20% and it was also estimated that the NRM, corrected for the parasitic magnetization, was similarly reduced. For the reasons mentioned above, all specimens were cleaned with HCl beforehand. The magnetization direction obtained after such a treatment was regarded as being the NRM.

5.3. Results

(1) The progressive cleaning by alternating fields indicated that the NRM consisted of a weak component eliminated at about 16 mT and of a strong one that was considered the characteristic magnetization. A typical example of demagnetization curves is given in Fig. 4. Table 2 summarizes the characteristics of the magnetizations for each site, and Fig. 5 displays the variation of the magnetization intensity with the alternating field value. The curves $J/J_0 = f(H)$ allow the definition of a median destructive field $\tilde{H}_c$ as the peak value of the alternating field where the ratio J/J_0 is equal to $1/2$. Table 2 indicates the $\tilde{H}_c$ mean value and the range of the observed values. No particular correlation appears between the $\tilde{H}_c$ values and the types of demagnetization curves or characteristic directions.

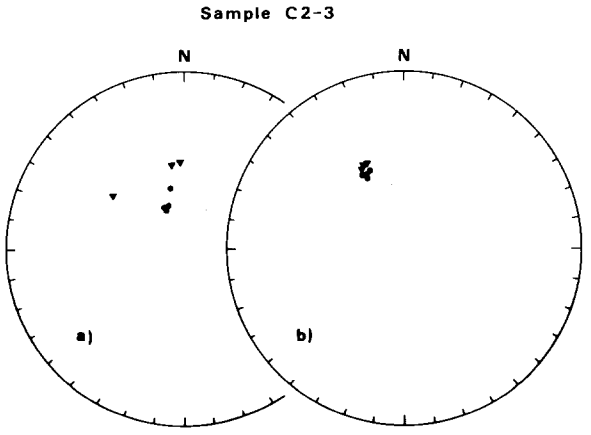


Fig. 2. Stereographic projections of NRM of specimens belonging to sample C2-3: (a) before chemical cleaning, (b) after chemical cleaning. ▼=cubes; ●=cores.

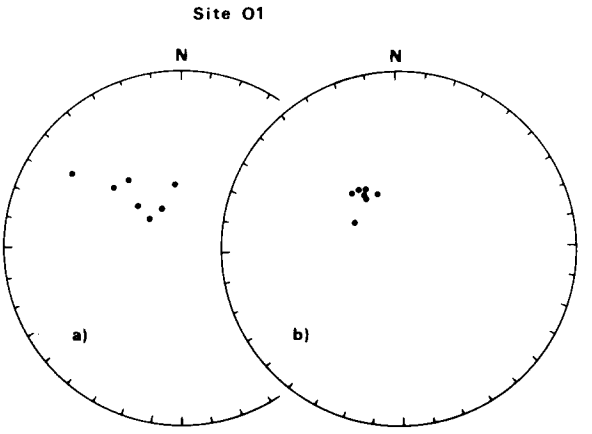
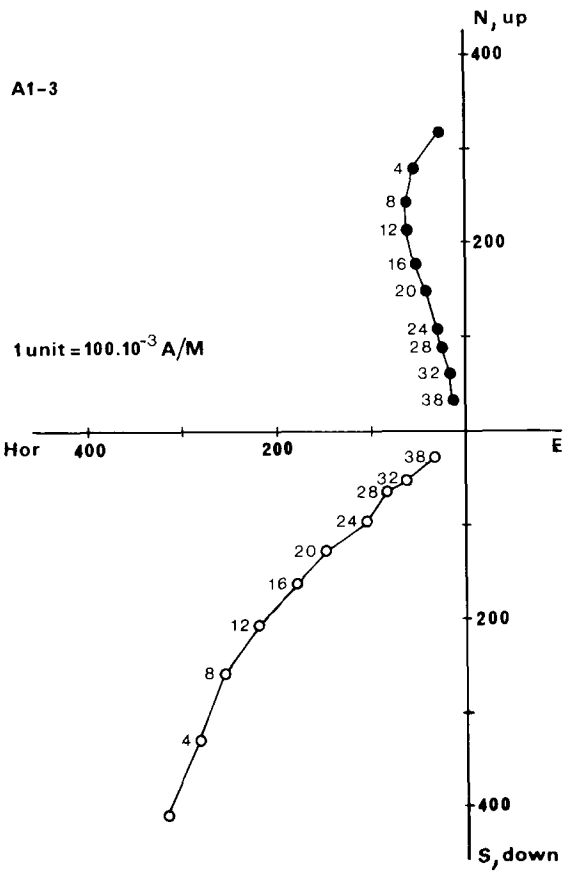


Fig. 3. Stereographic projections of NRM of samples from site O1: (a) before chemical cleaning, (b) after chemical cleaning.

(2) The variation of susceptibility curves in relation to temperature is rather simple and does not reveal any intermediate Curie point. Fig. 6 shows examples of such curves. Furthermore, the Hopkinson effect varies depending on the cases. In each case these curves are irreversible. We observe always an evolution of the cooling curve towards the decreasing temperatures. Some curves show evidence of partial oxidation during the heating and cooling cycle (see for example Fig. 6b).

Further cycles are by far more reversible. They give curves similar to the first cooling curve. The Curie points are determined on the cooling curves that correspond to a more stable state of magnetization. The results reveal the occurrence of Ti-poor minerals belonging to the titanomagnetite series with less than 10% ulvöspinel.



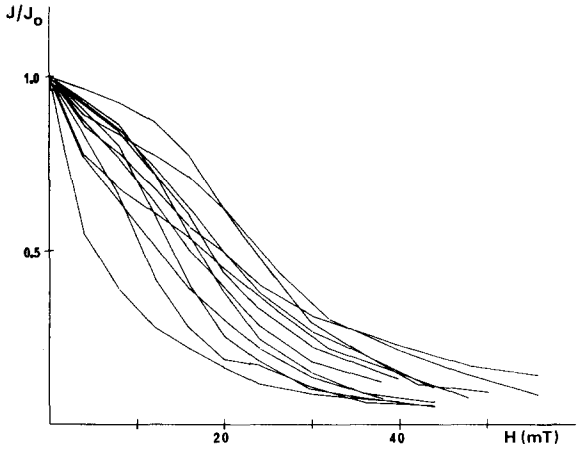


Fig. 5. Mean normalized a.c. magnetic field decay curves for each site. J_0 = intensity of NRM after chemical cleaning. Magnetic field (peak values) in milliteslas.

The characteristics of the mean anisotropy tensor of each site are summarized in Table 3. The mean direction of maximal and minimal suscep-

Fig. 4. Diagram of a progressive a.c. field demagnetization. Full symbols represent projections on the horizontal plane; open symbols represent projections on the vertical plane containing the horizontal component (denoted Hor). Numbers denote a.c. magnetic field peak values in milliteslas (mT).

TABLE 2

Site location and magnetic data

Site	Location		N	D (°)	I (°)	k_0	k	α_{95} (°)	J_0 ($\times 10^{-3}$ A/m)	Susceptibility χ	T_c (°C)	$\tilde{H}_c$ (mT)	VRM (%)
	lat.°N	long.°W											
A1	40.68	4.90	11	341.5	38.5	119	658	1.8	860±100	130±10	546	12.0–23.0	6±3
A2	40.53	5.17	8	349.5	49.0	65	64	7.0	1100±350	160±40	516	20.0–25.0	4±2
P2	40.05	6.05	11	357.5	41.0	77	250	2.9	730±150	140±10	513	10.0–12.0	6±3
P1	39.82	6.33	13	335.0	56.5	59	172	3.2	900±300	100±20	558	11.0–16.0	3±1
C1	39.65	6.52	12	344.5	44.0	96	194	3.1	2200±900	160±50	–	16.0–18.0	3±1
C2	39.55	6.63	11	337.0	32.0	72	232	3.0	1350±200	120±20	564	23.0–24.0	3±1
CM	39.07	7.02	16	328.5	54.0	57	130	3.2	1600±250	110±15	–	16.0–23.0	6±3
AL	38.68	7.36	8	355.5	52.0	41	89	5.9	750±300	120±30	573	11.0–13.0	5±2
POI	38.34	7.68	8	313.5	49.5	447	438	2.6	1100±100	115±30	–	16.0–25.0	3±1
POII	38.34	7.68	7	342.0	55.5	18	209	4.2	850±150	120±25	–	4.0– 7.0	4±1
O2	37.63	8.58	11	356.0	41.5	109	316	2.6	750±200	100±40	523	14.0–16.0	4±2
O1	37.57	8.66	14	324.0	42.5	84	383	2.0	750±200	90±10	–	20.0–26.0	6±2

N = number of specimens; D = mean characteristic magnetization declination; I = mean characteristic magnetization inclination, positive downward; k = precision parameter [57] for characteristic magnetization; k_0 = precision parameter after chemical cleaning and before demagnetization; α_{95} = semi-angle of 95% confidence cone. J_0 = mean NRM intensity; χ = mean volume susceptibility (dimensionless in the SI system); T_c = Curie temperature (not measured when not indicated); $\tilde{H}_c$ = interval of median destructive field (see text); VRM = viscous remanent magnetization ratio defined by Thellier's test [30] (see text).

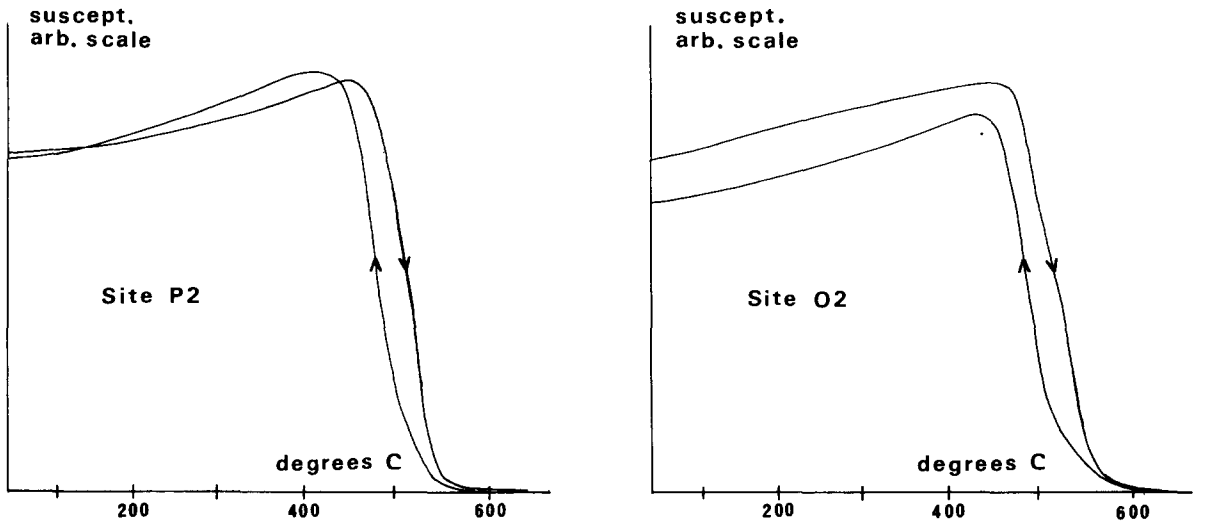


Fig. 6. Examples of thermal change in magnetic susceptibility, measured in a weak a.c. field.

tibility axes are indicated whenever they reveal a significant cluster as well as the θ_{63} angular variance. No lineation or foliation tendency is shown even within a given site. On the other hand, the anisotropy ratios are usually high, the peak value being 15%. The measured values are far above the 5% considered to be the upper limit consistent with the parallelism of the TRM and the ambient field [36]. Nevertheless, as pointed out by Uyeda et al. [37], a definite correlation between NRM direction and tensor of the susceptibility anisotropy is not well established. However, we tested the parallelism of the TRM direction and the ambient field. Ten cores with anisotropy ratios higher than 10% were heated up to 550°C and then cooled to room temperature in a given field with a 45° inclination; the minimal susceptibility axis was vertical.

TABLE 3

Magnetic anisotropy of susceptibility data

Site	N	Maximum susceptibility axis			Minimum susceptibility axis			Susceptibility
		D	I	θ_{63}	D	I	θ_{63}	ratio
A1	3	248	20	15	–	–	–	5.5±1.5
A2	2	–	–	–	–	–	–	13.0±6.5
P2	3	139	33	22	–	–	–	4.0±1.5
P1	5	–	–	–	310	11	20	6.0±2.5
C1	6	107	4	23	–	–	–	6.5±2.5
C2	4	217	18	31	–	–	–	11.0±2.0
CM	8	–	–	–	–	–	–	6.0±3.0
AL	4	–	–	–	313	6	16	6.5±2.5
POI	3	116	84	22	270	4	8	6.0±1.5
POII	4	–	–	–	302	6	15	6.0±2.5
O2	2	–	–	–	–	–	–	5.0±1.5
O1	5	346	63	24	258	2	18	9.5±2.0

N = number of specimens; θ_{63} = circular standard deviation; susceptibility ratio = $100\,(\chi_{max}-\chi_{min})/\bar{\chi}_m$, where $\bar{\chi}_m = \frac{1}{3}(\chi_{max}+\chi_{int}+\chi_{min})$.

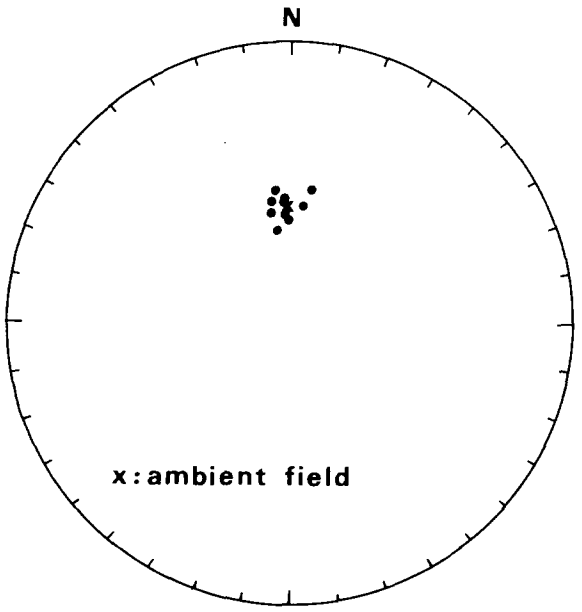


Fig. 7. Directions of TRM acquired on heating at 550°C by 10 samples with anisotropy of susceptibility ratios comprised between 10% and 15% (see Table 3 for definition).

In Fig. 7 one can find the TRM directions acquired by the specimens. No particular trend in direction appears and the average direction is at an angle of 0.8° to the applied field. This value is significantly lower than the confidence limit ($\alpha_{95} = 3.6°$). The alternating field stepwise demagnetization carried out on four specimens reveals that the demagnetization curves are quite similar to the NRM one's. We may, therefore, assume that in both cases the same magnetic grains carry the magnetization. The susceptibility anisotropy does not seem to cause any appreciable change in the TRM direction. Furthermore, if all sites are taken into account, this influence is still less sensitive, since the axes of the anisotropy tensors display large scattering.

6. Discussion

6.1. Significance of the characteristic directions

The NRM demagnetization curves with alternating field cleaning reveal that the NRM is rather simple after the removal of sawing and coring parasitic magnetization. It consists of a weak component, erased with a 16-mT alternating field and of a strong characteristic one. The small secondary magnetization cannot be determined. It would most likely correspond to the VRM acquired in the laboratory field. The characteristic magnetization is, therefore, assumed to have occurred at the time of dike emplacement. Part of the characteristic directions, though, are near the present geomagnetic field. Nevertheless, one cannot conclude that we are dealing with present directions since the angular discrepancy between the present field direction and the site mean direction is 12°, or equal to θ_{63}. The close proximity of present and ancient directions is not unlikely. Furthermore, the NRM of those particular sites (viscosity index, coercive field, demagnetization curves) are similar to those of the remaining sites. Therefore, no criterion exists to discard them. Only normal polarities have been determined. This result is consistent with the assumed ages of the characteristic magnetizations and the polarity time scale for the Mesozoic [58]. The radiometric ages given in Table 1 range between 205 and 160 Ma. The time span is well included in the so-called Graham normal interval. Three reverse events are mentioned in the normal interval, at 194, 180 and 168 Ma, but the rather discontinuous magnetic record of our study did not detect them.

The occurrence of a susceptibility anisotropy raises the question as to whether the TRM direction was parallel to the geomagnetic field direction when it was created. As shown previously, there exists no experimental clue giving rise to such a hypothesis, although the susceptibility anisotropy is uncommonly high.

All the site mean parameters are reported in Table 4. No site has been discarded in the computation, even those with a direction near the present geomagnetic field of the sampling area.

The time spans represented by the NRM at any given site are assumed to be short. The width sampled does not exceed hundred meters in any case. It appears from figs. 2 and 3 of Jaeger [59, pp. 312,313] that a sheet of this thickness would acquire its TRM over a period of approximately 200 years. This estimate supposes that the depth of

TABLE 4

Paleomagnetic poles for the Messejana dike

Site	Location		δ_p	δ_m
	lat.°N	long.°E		
A1	65.5	220.5	1.3	2.1
A2	76.5	217.0	6.1	9.2
P2	73.4	182.5	2.1	3.5
P1	70.5	264.0	3.3	4.6
C1	71.0	221.5	2.4	3.9
C2	60.5	222.0	1.9	3.4
CM	64.5	263.0	3.1	4.5
AL	83.0	204.5	5.5	8.1
PO1	51.0	265.0	2.3	3.4
PO2	75.5	259.0	4.3	6.0
O2	76.0	186.5	1.9	3.2
O1	57.0	249.0	1.5	2.5
Mean			k	α_{95}
pole	70.9°N	236.2°E	36	7.3

δ_m, δ_p = length of the axes of the ellipse of confidence at the 95% level, in degrees. Other symbols as in Table 2.

burial of the dike is greater than its thickness. The actual depth at the time of intrusion remains unknown. Thus 200 years can serve as a maximum estimate. The secular variation is, therefore, not precluded in the site mean directions. On the other hand, the mean pole corresponds to a larger time span if K-Ar dates are taken into account. The magmatic activity may have lasted over 40 Ma, that is, through the Early and Middle Jurassic times. As no definite correlation appears between K-Ar ages and the VGP positions, an Early to Middle Jurassic age is assumed for the mean pole. The value of the k parameter is 36, the 95% confidence limits being 18 and 60 [38]. Dalrymple et al. [2] obtained a similar dispersion parameter on the tholeiitic diabase and sills of Liberia. Since our paleolatitude is higher (27° instead of 7°), the k parameter should be lower for the Iberian dike if the models of secular variation hold true (see for instance [39]), but the theoretical variation is not significant at the 95% confidence level.

6.2. *Intercontinental comparison of geomagnetic poles*

We assume, following Schermerhorn et al. [8], that the emplacement of the Messejana dike is related to the magmatic activity that coincides with the beginning of the opening of the Atlantic Ocean. This leads us to compare the pole position obtained in the present study with those of similar formations in North America and Africa. Table 5 and Fig. 8 illustrate the similarity of the Iberian pole with the mean pole of African Mesozoic diabase in the present position of the continents. One could well conclude that from the Jurassic to the present time the whole movement of Africa and Iberia relative to Europe has been nearly identical. A similar direct comparison between Iberian and European directions of the same age would be very interesting. Unfortunately, there is an almost complete lack of data on European formations of Jurassic age. To circumvent this difficulty we can determine a European pole through those of North America and Africa by using one of the proposed reconstructions [40–44]. The criteria used to select the poles are those described by Beck [6] and Dalrymple et al. [2]. In this way, the secular variation may be eliminated as much as possible. Whenever a geological unit (dike, sill or lava flow) is represented by only one site, the VGP's are grouped geographically. This technique has been applied to the sills of the Draa Valley in Morocco, in Mauritania (sills and dikes) and in Liberia (dikes). The North Mountain VGP's have been rejected from the computation of the mean pole of North America because they are not closely connected to the other poles of this continent (the angular difference between the North American mean pole and those of North Mountain amounts to 21° for the upper lava flow and 12° for the lower one). One should note that the mean pole position is not significantly changed but the value of the k parameter shifts from 82 to 123.

For a similar reason the Sierra Leone poles have been discarded in the computation of the mean pole of Africa. Furthermore, the number of samples studied was very small.

The rejection of these two data enhances the value of the precision parameter from 115 to 209. The poles from Morocco, north of the South Atlas fault, have not been taken in consideration because the relations between Morocco and the African plate are not well established [45]. Like-

TABLE 5

Paleomagnetic poles used in continental reconstruction

Locality and rock type	Number of samples/ sites/and rock units	Location		α_{95}	References	K-Ar ages (Ma)	References
		lat. °N	long. °E				
Eastern North America							
Nova Scotia							
North Mountain basalt							
upper lavas	21/16/?	71	149	4.7	[51]	217	[51]
lower lavas	17/11/?	67	67	5.5		178, 196, 204	
Shelburne dike	11/2/1	69	98	4.5	[52]	197	[52]
Massachussetts							
Holyoke basalt and Grandby tuffs	16/5/5?	55.5	91.1	12.0	[53]		
Connecticut							
Talcott basalt	51/10/2	51	78	4.0	[5]		[5]
Holyoke basalt	81/10/2	60	83	3.0		193,161,176,197,201	[54]
Hampden basalt	76/14/9	73	92	4.0		195	[54]
West rock sill	18/3/1	63	89	4.0		190,198,201	[54]
Mount Carmel sill	33/5/1	61	87	4.0			
Buttress dike, Cross Rock dike, Fairhaven dike, Bridgeport dike, Higganum dike	?/41/5	69.5	100	4.5	[7]		
New Jersey							
Watchung basalt							
Palisades, Sourland Mountain; Belle Mountain, Balpate Mountain, sills	14/5/4	66.0	115.3	9.6	[55]	190,193,202,202 (Palisades)	[56]
						192	[54]
Pennsylvania and Maryland							
Quatertown sill	50/9/1	66	98.5	4.3	[6]		
Birdsboro sill	119/26/1	62	105	3.2			
Yorkhaven sill	173/37/1	61	105.5	1.5			
Gettysburg sill	56/10/1	63.2	101.5	5.7	[6,7]		
Boyds sill	44/7/1	67.3	107.4	3.4	[7]		
Frederick dikes	152/10/3	69.0	103.7	3.0			
Mean pole for Eastern North America	$N = 16$	64.0	95.2	3.3	$k = 123$		
Western Africa							
Morocco							
Draa Valley sills, MC1 to MC16	96/16/?	65.5	230.5	3.5	[3]	180–187	[3]
Mauretania (Hank)							
Touirist sill	33/5/1	67.5	220.6	3.6	(Sichler et al., in preparation)		
Azlaf sill	33/5/1	72.2	236.3	5.4			
Dikes	20/3/3	66.4	244.8	20.6			

Table 5 (continued)

Locality and rock type	Number of samples/ sites/and rock units	Location lat. °N	Location long. °E	α_{95}	References	K-Ar ages (Ma)	References
Mauretania (Hodh)							
Imourj massif	32/5/1	73.9	243.6	10.1	(Sichler et al., in preparation)	197,192	(Sichler et al., in prepration)
Bou Rzama sill	7/1/1	72.4	250.9	–		200,206,176	
Dikes	38/7/?	69.5	237.0	10.5		187,187,192	
Sierra Leone							
Freetown Complex							
normal polarity	7/7/1	78.2	221.3	5.5	[1]	165,172,194	[1]
reversed polarity	3/3/1	87.0	254.5	8.4			
Liberia							
Dikes, Liberian age province	124/12/11	67.5	246.7	7.1	[2]	173,176,178,188	[2]
Dikes and sills, Paneysville Sandstone and Pan-african age province:							
normal polarity	74/7/7	74.4	221.1	10.4			
reversed polarity	59/6/6	62.0	249.4	13.0		191,198	
Mean pole for Western Africa	N =10	69.4	238.5	3.3	k = 209		

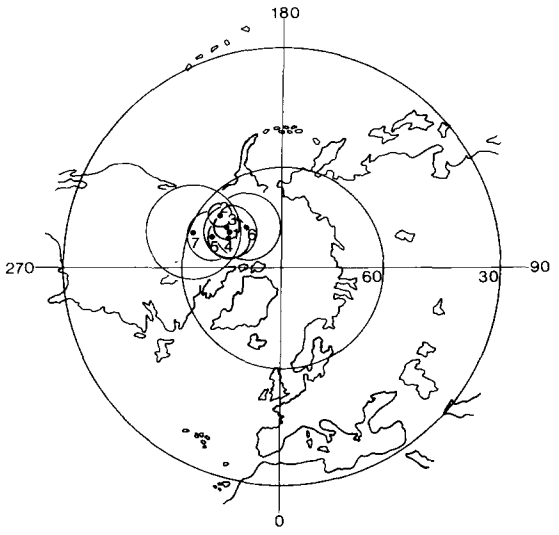


Fig. 8. Poles for Africa and Iberia, drawn with 95% confidence circles. Map is stereographic projection centered on north pole. *1* = this study, *2* = Draa Valley sills, *3* = Hank (Mauritania), *4* = Hodh (Mauritania), *5* = Liberia (Liberian age province), *6* = Liberia (Panafrican age province, normal polarity), *7* = Liberia (Panafrican age province, reversed polarity).

wise, the reconstructions that minimize the angular difference between the poles of Africa and North America were taken into account. Table 6 presents the data inferred from these various reconstructions. Among them the most pertinent are those of Bullard et al. [40], Le Pichon et al. [42] and Bonnin [44].

The angular difference between the European poles obtained from these three reconstructions is 6°, that is to say, less than the uncertainty affecting the positions of the North American and African poles and the parameters of rotation (pole positions, angles). Fig. 9 depicts the magnetization directions yielded in Madrid by each of the poles determined in this study and also those of stable Europe presented in Table 6. The inclinations are quite similar and the differences between the mean declinations are 35.5° in the reconstruction of Bullard et al., 42° in that of Le Pichon et al. and 39.5° in that of Bonnin. The comparison of Lower Permian results from Europe and Spain led to

TABLE 6

Calculation of an European pole for the Upper Triassic/Lower Jurassic time (convention for rotation poles as in Le Pichon et al. [42]; Europe in its present position)

Reference	Rotation pole NAM/EUR			Rotation pole AFR/EUR			Pole NAM after rotation		Pole AFR after rotation		$\delta^{*}_{NAM/AFR}$	Pole EUR**	
	lat.°N	long.°E	angle	lat.°N	long.°E	angle	lat.°N	long.°E	lat.°N	long.°E		lat.°N	long.°E
[40]	88.5	27.7	38.0	50.0	3.2	−41.4	65.0	133.1	69.7	128.8	5.3	67.6	131.3
[41]	88.4	27.4	38.0	44.6	−0.8	−50.2	65.0	133.1	63.9	108.1	10.7		
[42]	85.0	129.9	26.7	52.55	−0.4	−49.6	63.1	117.6	67.4	114.9	4.7	65.4	116.4
[43]	88.5	26.8	38.1	58.7	0.34	−33.1	65.0	133.2	75.1	157.0	13.2		
[44]	89.1	−103.1	29.6	53.9	−1.45	−48.3	63.7	125.6	68.9	117.2	6.6	66.6	122.0

* $\delta_{NAM/AFR}$ = angular distance between North American and African poles after rotation.

**Pole EUR = mean pole from pole NAM and pole AFR after rotation, calculated only from the reconstruction giving the smallest $\delta_{NAM/AFR}$.

similar conclusions, the difference between declinations being 35° [46]. One should also point out, as displayed in Fig. 9, the satisfactory accordance between the European directions deduced from the miscellaneous reconstructions and those inferred from the North Pyrenean VGP obtained by Girdler [47]. The angular difference of 35° in declination found for the Lower Permian times is thus confirmed by this study. This implies that no significant rotation of the Iberian plate could have taken place before the Middle Jurassic as some authors proposed [48–50]. They assumed that the first rotation took place during the Triassic. Their hypothesis was based mainly on sedimentological and paleogeographical evidence. The similitude of the Iberian pole with the African poles means that the two plates have undergone quite parallel movements since the Middle Jurassic. This statement agrees with models such as that of Le Pichon et al., but there remains the question of the magnitude of the rotation. The value revealed by the paleomagnetic studies is significantly greater than those presumed in their model. This discrepancy raises the problem of the initial fit between the Iberian and the European plates.

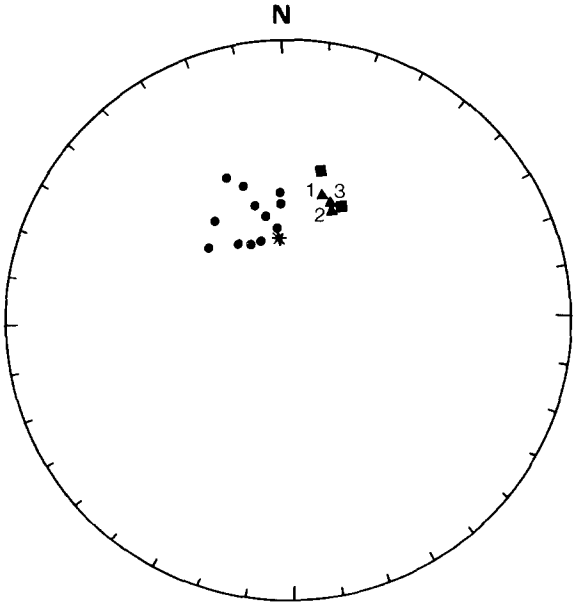


Fig. 9. Stereographic projections of the magnetic directions for the present-day location of Madrid, computed from Late Triassic/Early Jurassic pole position. ● = Iberia, this study; ■ = North Pyrenees [47]; ▲ = stable Europe, computed from Table 5. Asterisk denotes present-day geomagnetic field direction at Madrid.

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